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None
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# EMBEDDED REPOSITORY MANIFEST BLOCK (GOV-05 / MAN-001 COMPLIANT)

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baseline_release_tag: "v1.3.0-exec"

synchronization_hash: "A-77-DELTA-SHIELD-LOCKED"

sovereign_constancy: "C_ops == 0 [INVARIANT_CLAMP]"

temporal_override_floor: "tau_override <= 11.99ms
[INVARIANT_CLAMP]"

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ARC-FAQ-2026-PHASE7-01 | Version v1.3.0-exec | STATUS: FROZEN

Phase 7 Ecosystem Frequently Asked Questions & Living
Architecture FAQ

Anchor: A-77-DELTA-SHIELD-LOCKED | Effective: Sep 04, 2026

DOCUMENT METADATA

Title: Phase 7 Ecosystem Frequently Asked Questions & Living
Architecture FAQ

Spec ID: ARC-FAQ-2026-PHASE7-01

Version: 7.5 - REV-12 Reconciled Master Baseline

Target Substrate: Master Substrate – Phase 7, Phase 9 & Phase 10
Hardware Baseline (ARC-SPEC-2026-HWBL-07 REV-02)

Parent System: Subsystem 9 – Open Protocol & Boundary Protection
Specifications

Security Level: Public Reference & Executive Architecture FAQ

Cryptographic Hash Anchor: A-77-DELTA-SHIELD-LOCKED

Formal Audit Reference: ARC-AUDIT-2026-V14.6

Master Index Reference: ARC-SPEC-2026-MSAB-01 (REV-02)

1. EXECUTIVE ARCHITECTURE OVERVIEW & SYSTEM INTENT

Q1: What is the primary purpose of the Arcstone Computational Spine (ACS) Phase 7 Baseline?

A1: The Phase 7 Hardware Baseline provides a deterministic, fail-closed runtime substrate that bridges bare-metal silicon enforcement with software-based capability isolation. It establishes an "authorization-first" model where computational capability can never exceed granted institutional authority or structural safety boundaries, decoupling probabilistic software cognition (e.g., AI planners, heuristic engines) from physical safety authority.

Q2: How does Phase 7 differ from traditional permissive execution models?

A2: Traditional computing uses a permissive execution model where a system generates and executes an action first, relying on post-hoc sandboxing, monitoring, or kill-switches to catch errors. Phase 7 reframes execution around a single invariant: Existence != Permission. All computational workloads, state mutations, model-generated GPU kernels, public e-commerce order payloads, and physical mechatronic toolpaths must undergo deterministic pre-execution admissibility evaluation before hardware resources, stepper clocks, or energy reserves are allocated.

Q2b: What does Sovereign Constancy ($C_{ops} == 0$) mean in practice?

A2b: $C_{ops} == 0$ defines Zero Operational Drag / Zero Founder Drag. The system operates on self-enforcing, deterministic invariants

requiring zero synchronous human maintenance, live debugging meetings, or runtime operational SLAs. All authority transitions and policy updates flow asynchronously via signed cryptographic proofs.

2. CORE ARCHITECTURAL PILLARS, INVARIANTS & GOVERNANCE CONTROLS

Q3: What is the Dual-Baseline Topology?

A3: Phase 7 employs a dual-baseline architecture that creates an un-collapsible separation between physical hardware primitives and software execution domains:

- Physical Hardware Baseline: Built on TPM 2.0 roots of trust, Secure Enclaves, IOMMU DMA hardware isolation, non-cacheable Tightly-Coupled Memory (TCM), and hardware monotonic counters.
- Software Capability Baseline: Enforces seL4-style CSpaces, isolated IPC domains, Generational Nonce Mapping Bridges (GNMB), static CapDL zero-dynamic-retyping memory models (ARC-SPEC-2026-SEL4-01), and deterministic execution membranes.

Q4: How do Subsystem 9 specifications enforce Sovereign Systems over "False Systems"?

A4: Under ARC-SPEC-2026-GOV-02 and GOV-05, a system lacking machine-enforceable grammar is classified as a False System—an architecture reliant on human code reviews, manual bug-fixing, and developer promises that inevitably succumbs to Architecture Drift. A Sovereign System embeds non-negotiable rules directly into foundational types via the Principal Invariant Architect (PIA) mandate, ARC-SPEC-2026-CAF-01 4-tier downward governance (Tier I > Tier II > Tier III > Tier IV), ARC-SPEC-2026-REAL-02 downstream

authority prohibitions (L_upstream -> L_downstream),
ARC-SPEC-2026-FS-01 structural graph auditing, and the 8-Tier Audit
Matrix (AUD-01 to AUD-08 in ARC-RCD-SPEC-001):

- Machine-Enforced Grammar: Breaking a rule is physically impossible at compile-time (C_ops = 0).
- Spec-as-Code Control Plane: Technical Specification Documents (TSDs) exist as version-controlled YAML sidecars (/docs/specs/SPEC-xxx.yaml and embedded manifest.yaml blocks).
- Single-Source manifest.yaml & ComponentID: Every component generates a deterministic unique ComponentID matching $^{[A-Z0-9]^+-[A-Z0-9]\{8\}}$ derived from the initial git commit hash.
- 8-Tier Audit Matrix (AUD-01 to AUD-08): Enforces input hardening (AUD-01), zero varints on synchronous paths (AUD-02), Kingman queue saturation limits (AUD-05), append-only DAG provenance (omega_split, omega_merge) (AUD-06), and the out-of-band 3-of-5 Cryptographic PKCS#11 HSM Quorum Thaw protocol (AUD-08 / ARC-SPEC-2026-AUD-08).
- Auto-Discovery & PR Gates: Local arcstone-lint pre-commit tools and central CI/CD pipelines (ARC-SPEC-2026-OPS-01 / substrate-ci.yaml) enforce schema completeness, tag casing, and layer-boundary DAG directionality, automatically blocking PR merges on critical drift (POSIX 40).
- 48-Hour Break-Glass Auto-Downgrade: Emergency bypasses (--break-glass) emit audited tracking payloads. If un-rectified after 48 hours, an automated cron job auto-downgrades the component status back to Draft.

Q5: How does the State Engine ($Z = X \times C \times M$) and Dominance Lattice enforce fail-closed integrity?

A5: In accordance with ARC-RCD-SPEC-002, state transitions operate across a coupled state vector $Z = X \times C \times M$ (Operational State X , Constraint Config C , Meta-Constraints M) under three-tier evaluation masks ($IsAllowed_U$, $IsAllowed_V$, $IsAllowed_W$). System verification executes via a multi-stage decoupled architecture:

- Phase 1: Wire-Speed Driver Ingress Proxy (0(1) Hardware Shield - Gate 1): Executed directly inside the network interface driver via eBPF XDP (`edge_ingress_xdp.c`) prior to socket allocation. Frames breaching the 4096-byte (4KB) raw wire payload limit drop instantly at the NIC hardware level (`XDP_DROP` / `POSIX 30`).

- Phase 2: SRAM Dual-Buffer Staging & Security Filtering (Gate 2 - RFC-2026-SPINE-03 / ADR-003-REV-03): Executes single-pass $O(N)$ JCS canonicalization from static SRAM `buffer_a_raw` to `buffer_b_canonical`. Post-canonicalization expansion exceeding 4096 bytes triggers an atomic write-head halt, volatile memory zeroization (`volatile_purge_frame`), and exits via `POSIX 30` (`LEDGER_CORRUPTION`). Concurrently, parallel hardware security engines scan for adversarial prompt overrides; if detected, `POSIX 40` (`SECURITY_BREACH`) takes absolute precedence over `POSIX 30` drops.

- Phase 3: In-Band Canonical Model Dominance Poset: Validated frames pass to the non-collapsible dominance lattice:

`FAIL (POSIX 40) > FREEZE (POSIX 10) > PWC (POSIX 10) > REFUSAL (POSIX 32) > PASS (POSIX 0)`

Higher control planes maintain absolute operational precedence over downstream optimizers, neural networks, or motion planners. A single FAIL condition across any layer triggers an immediate global shutdown, bare-metal circuit-breaker lockout (`POSIX 40` via `hwbl_nmi_lockout.c`), or physical E-stop loop. Operational ambiguity, evidence insufficiency, vector clock drift (>50 ticks), or sensor degradation automatically forces a synchronous state freeze (`POSIX 10` sub-status `0x10B0`) with zero operational drag (`C_ops = 0`).

Q5b: How does the Generational Nonce Mapping Bridge (GNMB) interface prevent temporal logic collisions without breaching the sub-12ms floor?

A5b: ARC-SPEC-2026-GNMB-01 and ADR-003-REV-03 decouple long-term discrete output models from synchronous control-plane verification loops.

- Protocol Bipartitioning: Protobuf wire schemas (GenerationalStateNonce) are restricted strictly to asynchronous downstream telemetry and audit logging. Variable varint encoding is banned on the synchronous critical path (AUD-02).
- Synchronous Fixed Framing: Synchronous kernel verification requires a zero-copy, fixed-width frame (GnmbSyncCommitmentFrame) matching the packed 64-byte Little-Endian CfsrHeader layout (cfsr_ingest.rs). HMAC signature checks complete in $\leq 0.45\text{ms}$ in TCM, well inside the hard temporal ceiling ($\tau_{\text{override}} \leq 11.99\text{ms}$). Payloads exceeding the 4096-byte (4KB) DMA ceiling short-circuit instantly at Gate 1 eBPF pre-filtering, emitting POSIX 30 (LEDGER_CORRUPTION).

Q6: How are queue saturation and Kingman delay bounds managed under ARC-SPEC-2026-QDX-01, RFC-2026-SPINE-04, and Subsystem 9?

A6: ARC-SPEC-2026-QDX-01 incorporates Kingman's formula (W_q) to eliminate queue saturation asymptotes (AUD-05). System utilization is clamped at $\rho \leq 0.80$. Between the soft high-water mark ($\rho_{\text{soft}} = 0.80$) and the critical ceiling ($\rho_{\text{hard}} = 0.95$), Proportional Windowing Control (PWC) engages inside the Asynchronous Reconciliation Queue (ARC-SPEC-2026-TRQ-01):

- Incoming transaction ID hashes are evaluated against a fixed-point bitwise execution mask.

- If a hash collision occurs, the frame is assigned POSIX 10 (Sub-Status 0x10A0 PWC Throttled Hold), causing real-time physical actuators (ARC-SPEC-2026-HIL-01) to step down to a standby configuration (150°C hotend / zero-flow rate) without triggering limit-cycle oscillations.
- Out-of-band user-space metrics daemons (edge_metrics_daemon.rs in OPS-01 Appendix F) stream integer queue occupancy (rho_pct) into eBPF maps via bpf_map_update_elem. If queue saturation reaches absolute limits (rho >= 0.95), eBPF XDP filters drop packets directly at wire speed via XDP_DROP (POSIX 32 Refusal, sub-status 0x0020).

3. OFFLINE AI, ADDITIVE SYNTHESIS & ABSOLUTE LOCAL CONTINUITY

Q9: How does Phase 7 enable high-stakes AI workloads and physical fabrication (AM/3D Printing) to run safely in offline or air-gapped environments?

A9: Phase 7 achieves Absolute Local Continuity without requiring live calls to central cloud infrastructure or network identity providers by decoupling authority from transport. Safety and physical toolpath integrity are guaranteed locally via self-contained evaluation logic, local hardware roots of trust (TPM 2.0), pre-allocated invariant gates, and signed offline build archives.

Q10: What are the key technical mechanisms driving offline operation and physical synthesis?

A10:

- Admissible Computational Spines (ACS): Self-contained hardware/software cores carrying local evaluation rules and memory.
- Formally Verified seL4 Substrate (ARC-SPEC-2026-SEL4-01): Microkernel binary formally proven via Isabelle/HOL to operate under zero-dynamic-retyping and zero-heap runtime constraints.
- Root Task Privilege Shedding (ARC-SPEC-2026-SEL4-01): Revokes Root Task administrative handles post-boot via CapDL/CDT, shedding ambient authority before operational execution.
- Admissibility Kernel (AAK) & AAC-CERT (ARC-SPEC-2026-AAK-01): Upstream driver-level pre-execution filter for GPUs/TPUs using signed proof certificates (AAC-CERT JSON schema validated via Jest harness `validateAacCertVerification.ts`).
- Bare-Metal NMI Emergency Lockout (`hwbl_nmi_lockout.c`): C-kernel module intercepting vector clock drift breaches (>50 ticks) directly on silicon in $<100\mu\text{s}$. Flushes CPU cache (`wbinvd`), cuts hardware power relays via Priority 0 bus preemption, and shifts state via atomic static TCM bitset joins ($S_{\{t+1\}} = S_t \cup \Delta v$).
- 3-of-5 HSM Quorum Thaw Protocol (ARC-SPEC-2026-AUD-08): Air-gapped PKCS#11 HSM key ceremony required to unseal and re-baseline frozen contexts offloaded to TRQ-01 static ring buffers.
- Slicer-as-Compiler Locking (FAB-01): Packages G-code toolpaths inside signed .3mf containers with SHA-256 Slicer Binary Locks.
- Hardware-in-the-Loop Interlock Membrane (ARC-SPEC-2026-HIL-01): C++20 mechatronic membrane (`hil_actuator_membrane.hpp`) mapping POSIX signals directly to physical actuator safe-states in $\leq 11.99\text{ms}$.
- Capacitor-Backed FRAM State-Save (FAB-01): Sub-5ms emergency delta write during 15–25ms AC zero-cross hold-up window.

4. BARE-METAL ENFORCEMENT, CSS-01 REASON CODES & POSIX SIGNAL MAPPING

Q11: What bare-metal microarchitectural mechanisms enforce state isolation during frame evaluation and mechatronic motion?

A11: Phase 7 hardware execution membranes rely on six low-level microarchitectural enforcement primitives:

- eBPF XDP Wire-Speed Ingress Filter (OPS-01 Appendix E): Gate 1 drops oversized network packets (>4096B) directly at the NIC driver layer before allocating user-space socket buffers.
- TOCTOU Private-Copy Protection (ARC-SPEC-2026-SEL4-01): Microkernel transport membranes copy incoming shared-memory frames directly into private, non-cacheable Tightly-Coupled Memory (TCM) before evaluating invariants.
- Speculative Write Buffering (SWB): Store instructions route to an isolated SWB; writes cannot reach L1 cache or registers prior to validation by Pi(S). Rejection triggers a zero-branch SWB flush.
- Two-Pass Atomic Cleanup Protocol: Revocation of memory capabilities triggers an inline assembly mask barrier initiating a two-pass atomic purge.
- Dual-Gate Short-Circuit Order (SPINE-04 / EDGE-01 / HWBL-07 / ADR-003-REV-03): Raw wire byte boundaries ($\leq 4\text{KB}$) are verified by Gate 1 eBPF before JCS parsing; Gate 2 catches post-canonicalization expansions exceeding 4096 bytes inside transient staging SRAM before penetrating TCM (POSIX 30). Volatile memory purges clear all staging registers on drop.
- Host/MCU Determinism Split (FAB-01): Isolates non-deterministic trajectory generation (PREEMPT_RT Linux SBC) from hard real-time step scheduling (bare-metal 32-bit ARM MCU) over differential CAN-FD / RS-485.

Q12: How are fail-closed verdicts mapped to canonical POSIX signals and CSS-01 Reason Codes?

A12:

- POSIX 0 (PASS): Nominal Execution (GNMB_STATUS_PASS). Authorizes host execution space ingestion for sanitized payload frames or signed toolpaths ($\tau < 11.00\text{ms}$). Sub-status 0x0000.

- POSIX 10 (FREEZE / PWC): Disambiguated by hardware sub-status registers:

- Sub-Status 0x10A0 (PWC_THROTTLE_HOLD): Throttled Hold ($\rho \geq 0.80$) / Soft Yield ($11.00\text{ms} \leq \tau < 11.99\text{ms}$). Downstream real-time actuators step down to a standby configuration (150°C hotend / zero material flow rate).

- Sub-Status 0x10B0 (HARDWARE_CLOCK_FREEZE): Non-Maskable Interrupt temporal drift override stasis or non-fatal clock skew ($\tau < 11.99\text{ms}$). Commands physical actuators to drop stepper motor torque to zero. Context offloaded to TRQ-01 for out-of-band 3-of-5 Cryptographic HSM Quorum Thaw (AUD-08).

- POSIX 30 (LEDGER_CORRUPTION): Out-of-Band Edge Drop / Gate 1 Raw Payload > 4KB DMA Ceiling Breach / Gate 2 JCS Staging Expansion Breach (byte 4097 in SRAM Buffer B) / Structural Graph Contradiction (TC-01). Triggers immediate O(1) short-circuit and volatile memory purge (0x001E / 0x0302 ERR_STANDING_LEAK).

- POSIX 32 (REFUSAL): In-Band Policy Refusal / Front-Edge Queue Saturation Overflow ($\rho \geq 0.95$ via eBPF edge_ingress_xdp.c, sub-status 0x0020) / Kinematic Acceleration Violation / Inter-context Authority Leakage mapped to CSS-01 reason code 0x0301 (ERR_AUTHORITY_LEAK).

- POSIX 40 (SECURITY_BREACH): Invariant Violation / Invalid Signature or Nonce Breach / Vector Clock Slip >50 ticks /

Power-Loss Hold-Up / Unabsorbed Temporal Breach ($\tau \geq 11.99\text{ms}$, sub-status 0x0028) / Un-Contracted State Mutation (0x0302 ERR_STANDING_LEAK) / Log Visibility Breach (0x0303 ERR_VISIBILITY_LEAK) / Temporal Drift Breach (0x0304 ERR_TEMPORAL_DRIFT_LEAK) / Cognitive Variable Injection / Parallel Security Pattern Match / Unrecoverable RECURSIVE_COLLAPSE. Executed by hwbl_nmi_lockout.c: flushes volatile cache (wbinvd), drops hardware power relays via Priority 0 bus preemption, and forces unrecoverable silicon panic (C_ops=0). Overrides POSIX 30 drops.

5. CANONICAL COMPUTATIONAL SPINE & SUBSYSTEM 9 REGISTRY

Q13: What is the complete Suite of Formalized Subsystem 9 Specifications & Protocol Interfaces?

A13: Subsystem 9 unifies the complete open protocol and boundary protection suite governing perimeter protection, asynchronous queuing, integration testing, hardware interlocks, cryptographic thawing, extraterrestrial spatial gating, seL4 microkernel formal verification, and CI/CD orchestration:

- ARC-SPEC-2026-EDGE-01: Short-Circuit Perimeter Pre-Filtering Pipeline Specification.
- ARC-SPEC-2026-EDGE-02: Serverless Perimeter Gateway & POSIX Exit Code Specification.
- ARC-SPEC-2026-CFSR-02: Graded Commitment & Dynamic Action Cancellation Specification (REV 3.4).
- ARC-SPEC-2026-NAV-08: Architectural Normalization & Capability Progression Placement Specification.
- ARC-SPEC-2026-AFK-01: Admissibility-First Kernel Specification.

- ARC-SPEC-2026-TRQ-01: Asynchronous Reconciliation Queue Specification.
- ARC-SPEC-2026-AAK-01: Automated AAC-CERT Integration Test Specification & Jest Runner (validateAacCertVerification.ts).
- ARC-SPEC-2026-HIL-01: Hardware-in-the-Loop Acceptance Matrix Specification (hil_actuator_membrane.hpp).
- ARC-SPEC-2026-AUD-08: Cryptographic 3-of-5 HSM Quorum Thaw & Reconciliation Protocol (aud_reconciliation.rs).
- ARC-SPEC-2026-SADG-01: Extraterrestrial Spatial Admissibility & Deterministic Governance Kernel (E-SADGK v4.2).
- ARC-SPEC-2026-SEL4-01: High-Assurance System Architecture & sel4-Substrate Integration Specification (v1.3.0-exec).
- ARC-SPEC-2026-OPS-01 (REV-02): GitHub Actions CI/CD Deployment Orchestration Pipeline Specification (substrate-ci.yml) including:
 - Appendix E: Edge XDP Raw Ingress Pre-Filter Shield (edge_ingress_xdp.c).
 - Appendix F: User-Space eBPF Control Plane Metrics Daemon (edge_metrics_daemon.rs).
 - Appendix G: eBPF & Control Plane Cross-Compilation Script (build-ebpf-stack.sh).
 - Appendix H: Localized High-Load Packet Injection Simulator (mock_load_injector.py).
- ARC-SPEC-2026-GNMB-01 & ADR-003-REV-03: Generational Nonce Mapping Bridge & Canonical Remediation Binding (cfsr_ingest.rs).
- RFC-2026-SPINE-04 (REV-02): Predictive Epoch Windowing, Dynamic Temporal Shifting & Sub-Status Bitmask Mapping.

Q13b: How does ARC-SPEC-2026-CFSR-02 govern dynamic action canceling and commitment thresholds without breaching microarchitectural time limits?

A13b: ARC-SPEC-2026-CFSR-02 establishes a Graded Commitment Framework that decouples Layer 6 action initiation from mandatory recovery states across three normalized zones:

1. Startup / Feint Zone (0-25%): Aborting initiates a 250ms-400ms Global Feint Cooldown (ICD) to block high-APM macro exploits. Primary resources deducted at t_{input} are non-refundable.
2. Costed Cancel Zone (25-60%): Aborts require secondary stamina/meter expenditure; invincibility frames (I-frames) are suppressed until t_{commit} of the dodge itself.
3. Locked Commitment Point ($t_{commit} = 0.60$): Hard boundary beyond which actions cannot be canceled. Clamped to a physical minimum of 15.99ms to prevent clock-skew collisions under high-haste state modifiers.

Wire Mapping & Anti-Spam: All Layer 6 gameplay payload telemetry packs inside a 32-byte struct (`animation_payload_t`) passed into `IngressFrame->payload_buffer`, preserving the 64-byte `cfsr_header_t` layout. Input bursts breaching $I_{spam} > 3.5$ set canonical substatus Bit `0x0020` (`ERR_QUEUE_SATURATION_REFUSAL`) on `cfsr_header_t.substatus_mask`, forcing an immediate t_{commit} hard-lock and routing to TRQ under POSIX 32 without introducing operational drag ($C_{ops} = 0$).

Q13c: What is the architectural placement and governing boundary of ARC-SPEC-2026-NAV-08?

A13c: ARC-SPEC-2026-NAV-08 establishes the normalized placement of Document 08 (Capability Progression View) strictly within the non-canonical Navigation Layer.

- **Non-Constitutional Scope:** Classified permanently as DOCUMENT ONLY within /docs/navigation/. Independent repository admission is explicitly DENIED (per RAR-2026-09-03-DOC08). Scientific authority remains isolated within upstream constitutional repositories.

- **Canonical Nomenclature:** Enforces "Representative Computational Realization" (replacing draft "Potential Computational Spine") and "Observed Capability Gaps" (replacing draft "Candidate Capability Disciplines").

- **Dependency Pathway:** Downstream orientation follows the strict unidirectional pathway: Canonical Scientific Specs -> Foundational Realization States -> Capability Progression Mapping -> Atlas Integration (Document 05 v2.1).

Q13d: What is the purpose and line-rate execution constraint of ARC-SPEC-2026-AFK-01?

A13d: ARC-SPEC-2026-AFK-01 defines the Admissibility-First Kernel (AFK), an out-of-band, hardware-accelerated governance engine designed to prevent Silent Authority Inflation (SAI) in autonomous systems.

- **Line-Rate Execution:** Module D's line-rate admissibility gate operates out-of-band on eBPF ring-buffers/Smart-NIC microkernels with a worst-case execution time (WCET) budget $\leq 500\mu\text{s}$ at the 99.999th percentile, operating safely within the sub-12ms network floor ($\text{tau_override} \leq 11.99\text{ms}$).

- **Zero-Copy Shared Register:** Authority surfaces are mirrored into a packed 64-byte shared-memory structure (/dev/shm/afk_register) aligned to a single cache line using a Seqlock protocol, enabling zero-copy reads without Protobuf overhead or lock contention.

- **Contraction-Dominant Governance:** Normalizes pressure telemetry ($P_{\text{decoupled}} = \max(0, P_{\text{raw}} - P_{\text{self}})$) to scalar energy E in $[0.0,$

1.0], contracting admissible action surfaces across Nominal, Restricted, Critical, and FREEZE phases to uphold $C_{ops} = 0$.

6. MATTER-AS-A-SERVICE (MaaS) & AUTOMATED EDGE INGESTION

Q14: How do external storefronts and email requesters interact with off-planet manufacturing without introducing operational drag ($C_{ops} = 0$)?

A14:

- Cloudflare Edge Sanitizer (SIF-004 / ARC-SPEC-2026-EDGE-01): Inspects incoming requests at Layer 7. Payloads > 4KB exit POSIX 30; synchronous drag markers ("zoom call", "live demo") or broken criteria exit POSIX 40. Valid requests trigger automated Stripe invoicing settling directly into Mercury Bank with zero human drag.
- MaaS API Gateway (SPEC-MAAS-2026-API-1): Enables Shopify, SAP, or enterprise ERP applications to submit 3MF/STEP CAD files or bio UUIDs. Pre-flight physics verification runs in $O(1)$ time.
- Dual-Token Lock & Fallback Proxy Escrow: Orders issue Token A (Physical Build Token) and Token B (Professional Credential Token). If an expert's PKI hardware key is revoked mid-transit during down-well re-entry, the gateway executes Fallback Proxy Escrow Routing—temporarily pausing context (POSIX 10 0x10A0 PWC) and re-binding Token B to a pre-verified proxy in the target jurisdiction without cancelling the order.

7. FLAGPLANT GOVERNANCE & PATH ROUTING ARCHITECTURE

Q15: What is the Canonical Flagplant Policy for this repository?

A15: This repository is published as a Locked Canonical Reference Standard (v1.3.0-exec). In accordance with the Zero Founder Drag mandate (C_ops = 0), the core upstream specification tree is read-only. Upstream pull requests (PRs), issue queues, and direct mutations to invariant laws are disabled by design. Amendments and revisions flow strictly Upstream -> Downstream.

Q16: How do third-party developers and organizations interact with the substrate across Paths A, B, C, and D?

A16: Access, integration, and downstream deployment are strictly mapped across four governance tiers in accordance with LICENSE-SCOPE (ARC-LICENSE-SCOPE-2026-09-02), ARC-PAT-2026-CONV-ADDENDUM-V2.1, and ARC-SPEC-2026-SADG-01:

- Path A (Public Open Integration): Unrestricted forking for downstream applications and Docker simulation using sanitized JSON schemas (Open Access under Permissive Dual MIT / Apache-2.0).
- Path B (Commercial Substrate Deployment): Commercial spinouts, custom execution wrappers, enterprise CFN node deployments, MaaS storefront ingestion, and local AM print farm management (bound by ARC-SPEC-2026-REAL-01 and ARC-SPEC-2026-REAL-02).
- Path C (Institutional & Government Substrate): Terrestrial infrastructure, critical public utility backbones, autonomous industrial fabrication cells, and custom hardware enclave provisioning.
- Path D (Defense, Orbital & Air-Gapped Clearance): Deep-space bus chassis, orbital satellites, extraterrestrial habitat additive manufacturing, off-Earth habitat pressure and spatial gating (ARC-SPEC-2026-SADG-01 / OST Article II/VI/VIII compliance), formally verified seL4 microkernel enclaves (ARC-SPEC-2026-SEL4-01), and air-gapped defense networks enforcing Phase 10+ access.

Q17: How does a developer get started on Path A in under 10 minutes?

A17:

1. Clone & Environment Spin-Up: Run `git clone` and launch the environment using VS Code DevContainers or Docker (`docker-compose up`).
2. Deterministic Verification: Run `npm test` or `./scripts/build-ebpf-stack.sh` to execute local build and integration test suites (`tests/validateAacCertVerification.ts`) against A-77-DELTA-SHIELD-LOCKED.
3. Downstream Extension: Create a local branch (`git checkout -b feature/custom-membrane`) to build downstream realization wrappers.

[SYSTEM BLOCK LATCHED & FAQ SPECIFICATION SEALED
A-77-DELTA-SHIELD-LOCKED]